

What are the complications?

Last revised in December 2024

- **Microvascular complications** (due to damage to small blood vessels)
 - Nephropathy — diabetic kidney disease is caused by damage to small blood vessels in the kidneys.
 - Almost 20% of people with diabetes will need treatment for kidney disease during their lifetime [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - At least 10,350 people in the UK have end-stage kidney failure due to diabetes.
 - People with diabetes are 5 times more likely to need either kidney dialysis or a kidney transplant than people without diabetes.
 - Retinopathy — diabetic retinopathy is caused by small blood vessel damage to the retina, leading to progressive loss of vision and possible blindness [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Diabetes is responsible for 5% of all sight loss in the UK [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - Almost 50% of people with type 1 diabetes have some form of diabetic retinopathy; after 20 years of living with the disease, almost everyone with type 1 will have some degree of retinopathy.
 - About 14% of people of working age who have severe visual impairment have diabetes.
 - People with diabetes are more likely to get glaucoma and cataracts, both of which can lead to blindness.
 - Neuropathy — diabetes causes nerve damage through different mechanisms, including direct damage by the hyperglycemia and decreased blood flow to nerves by damaging small blood vessels [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Painful diabetic neuropathy is estimated to affect between 16–26% of people with diabetes [NICE, 2019c ([/topics/diabetes-type-1/references/](#))].
 - Diabetic foot disease often leads to ulceration and subsequent limb amputation [WHO, 2020 ([/topics/diabetes-type-1/references/](#))]. Between 70,000–90,000 people with diabetes have a foot ulcer in any given week, and diabetes leads to more than 8500 leg, toe, or foot amputations every year [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
 - Autonomic neuropathy is a late-stage complication of diabetes where the neurons of the autonomic nervous system become damaged owing to chronic hyperglycaemia. It presents in different ways and affects a variety of organs, including the skin (sweating), blood vessels (postural hypotension), gastrointestinal tract (gastroparesis and diarrhoea), heart, bladder function, and sexual function. It may also blunt the symptoms of [hypoglycaemia](#) ([/topics/diabetes-type-1/diagnosis/when-to-suspect-hypoglycaemia/](#)). [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].
 - Men with diabetes are 3 times more likely to have trouble getting or keeping an erection than men without diabetes. More than 25% of women with type 1 diabetes have sexual problems [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
- **Macrovascular complications** (due to damage to larger blood vessels)
 - Diabetes is a risk factor for the development of atherosclerosis, which increases the risk of cardiovascular diseases (CVDs), such as myocardial infarction, heart failure,

stroke, and peripheral arterial disease (for example intermittent claudication) [[WHO, 2020 \(/topics/diabetes-type-1/references/\)](#)].

- Type 1 diabetes is associated with a 2–4 fold increased risk of CVD in adults aged over 40 years, especially in the presence of renal disease [[ABCD-RA, 2017 \(/topics/diabetes-type-1/references/\)](#)].
- Compared with people without diabetes, people with type 1 diabetes are more than 4 times as likely to have a heart attack, 4.5 times more likely to experience heart failure, and 3.5 times more likely to have a stroke [[Diabetes UK, 2019 \(/topics/diabetes-type-1/references/\)](#)].

- **Metabolic complications**

- Diabetic ketoacidosis (DKA) is a serious complication of diabetes characterized by hyperglycaemia (or a diagnosis of diabetes), metabolic acidosis, and ketonaemia [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#); [Karslioglu French, 2019 \(/topics/diabetes-type-1/references/\)](#)].
 - Insulin deficiency results in an increase in counter-regulatory hormone production (glucagon, cortisol, growth hormone, and catecholamines). This type of hormonal imbalance enhances hepatic glycogenolysis (glucose production through the breakdown of glycogen stores) and gluconeogenesis (formation of glucose from other substrates), resulting in severe hyperglycaemia [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#)].
 - Enhanced lipolysis (due to the lack of utility of glucose) increases serum free fatty acids that are then metabolized as an alternative energy source in the process of ketogenesis, resulting in accumulation of large quantities of ketone bodies and subsequent metabolic acidosis [[Joint British Diabetes Societies Inpatient Care Group, 2013 \(/topics/diabetes-type-1/references/\)](#)].
 - DKA is a medical emergency because it leads to dehydration and electrolyte imbalances [[Karslioglu French, 2019 \(/topics/diabetes-type-1/references/\)](#)]. The mechanisms responsible for fluid depletion in DKA include osmotic diuresis due to hyperglycaemia, vomiting, and inability to take in fluid owing to a diminished level of consciousness. Electrolyte shifts and depletion are in part related to the osmotic diuresis.
- Hypoglycaemia is an inevitable adverse effect of insulin treatment. It is generally defined as being present when blood glucose levels fall to less than 3.5 mmol/L; however, the absolute blood glucose level at which signs and symptoms begin to occur can vary.
 - Severe hypoglycaemia may result in convulsions, inability to swallow, loss of consciousness, and coma. Between 4–10% of deaths in people with type 1 diabetes are attributed to hypoglycaemia [[NICE, 2015a \(/topics/diabetes-type-1/references/\)](#)].
 - See the section on [Hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](#) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](#) for more information on hypoglycaemia.

- **Other autoimmune conditions**

- People with type 1 diabetes are at increased risk of developing other autoimmune diseases, most commonly thyroid disease (Graves' disease or Hashimoto's thyroiditis, 15–30%), autoimmune gastritis and/or pernicious anaemia (5–10%), coeliac disease (4–9%), vitiligo (2–10%), and Addison's disease (0.5%) [[Van den Driessche, 2009 \(/topics/diabetes-type-1/references/\)](#)].

- **Psychological complications**

- Around 40% of people with diabetes experience emotional or psychological problems, such as anxiety, depression, and diabetes-related emotional distress (which affects about 42% of adults with type 1 diabetes). Poor emotional and psychological wellbeing is associated with poorer quality of life, greater difficulty with self management, and sub-optimal glycaemic control [Diabetes UK, 2018 ([/topics/diabetes-type-1/references/](#))].
- In children and young people, psychological complications may include behavioural and conduct disorders, family/relationship difficulties, and risk-taking behaviour (including non-adherence to recommended treatment) [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))].
- Needle phobias, fear of self-injecting, and eating disorders (which affect over 30% of women in their early twenties with type 1 diabetes) may lead to poor glycaemic control and subsequent complications [Diabetes UK, 2018 ([/topics/diabetes-type-1/references/](#))].
- A position statement (<http://diabetes-resources-production.s3-eu-west-1.amazonaws.com/diabetes-storage/2017-08/Diabulimia%20Position%20statement%20Mar%202017.doc>), issued by Diabetes UK states that around 40% of women and 11% of adolescent boys with diabetes may omit or reduce insulin to lose weight (diabulimia). Diabulimia is not a recognized medical or psychiatric term, but the Diagnostic Statistical Manual of Mental Disorders (DSM-5) considers that insulin omission in order to lose weight is a clinical feature of anorexia and bulimia.
- **Infections and other skin complications**
 - People with diabetes are prone to infections, particularly of the urinary tract and skin [Diabetes UK, 2019 ([/topics/diabetes-type-1/references/](#))], and severe infections can lead to amputations [WHO, 2020 ([/topics/diabetes-type-1/references/](#))].
 - Necrobiosis lipoidica is an uncommon inflammatory condition in which shiny, reddish-brown or yellowish patches develop in the skin of people with diabetes [BAD, 2019 ([/topics/diabetes-type-1/references/](#))].
- **Reduced quality of life**
 - People with diabetes face significant challenges to daily living, such as managing episodes of hypoglycaemia and hyperglycaemia, daily administration of insulin, regular self-monitoring of blood glucose, and the need to plan normal daily activities (such as eating and exercising) [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].
- **Reduced life expectancy**
 - Having type 1 diabetes typically reduces life expectancy in the UK by 11–15 years [NICE, 2015a ([/topics/diabetes-type-1/references/](#)); NICE, 2019b ([/topics/diabetes-type-1/references/](#))].
 - The risk of death is 135% higher than for people of the same age who do not have diabetes. Most of the deaths are due to chronic complications of diabetes, although death from acute hypoglycaemia or DKA may occur [NICE, 2015a ([/topics/diabetes-type-1/references/](#))].

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